GEOMETRY QUEST · WORLD 6

Polygons

ANSWER SHEET · for the grown-up

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

1 Name the polygon with this many sides.

5 sides	6 sides	8 sides	12 sides
pentagon	**hexagon**	**octagon**	**dodecagon**

2 Angle total = (sides − 2) × 180°. Find the total of the inside angles.

hexagon	octagon	decagon
720 °	**1080** °	**1440** °

3 Each inside angle of a regular polygon = angle total ÷ number of sides. Find one angle of a regular pentagon and a regular hexagon.

pentagon **108** ° **hexagon 120** °

4 True or false?

A square is a rectangle.	**true**
Every rectangle is a square.	**false**
A rhombus has four equal sides.	**true**
A trapezium has two pairs of parallel sides.	**false**

5 Find: (a) the area of a parallelogram with base 9 cm and height 4 cm; (b) the perimeter of a 7 cm by 3 cm rectangle.

(a) **36** cm² **(b) 20** cm

6 Polygon hunt: find three polygons at home or outside. Name each one and say where you found it.

Any sensible answers, e.g. hexagon: nut on a bolt.

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0